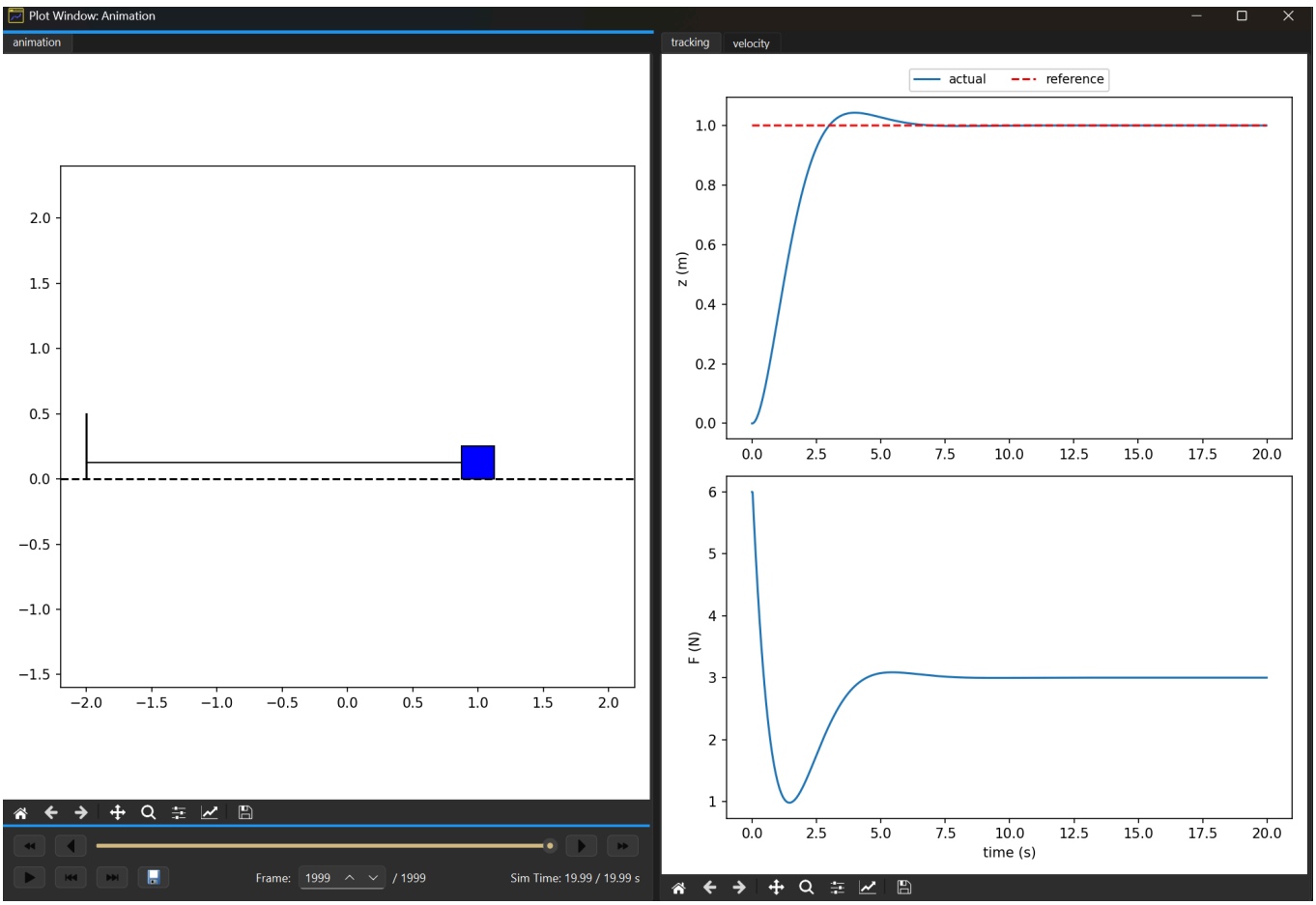
# Chapter 11

## D.11

```
des_poles = array([-0.7852+0.7854j, -0.7852-0.7854j])
self.K = array([[3.1666, 7.3516]])
self.kr = array([[6.1666]])
```

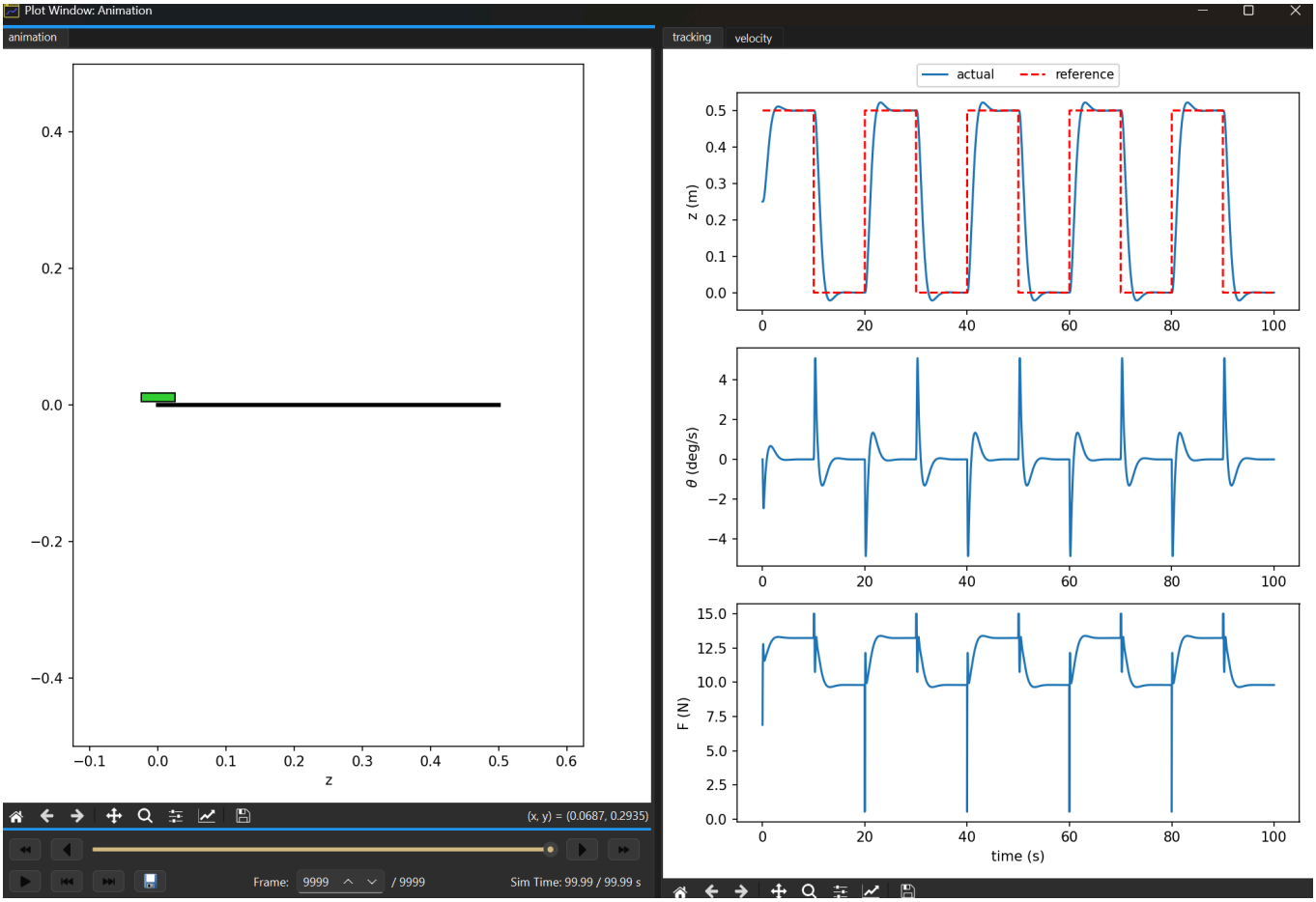
(d) It's the same gains because we're applying gains to the same states as PD control to get the same rise time and damping coefficient. There's no other way to do it.

(e) Didn't use digital derivative because we had permission not to from Dr. Killpack.



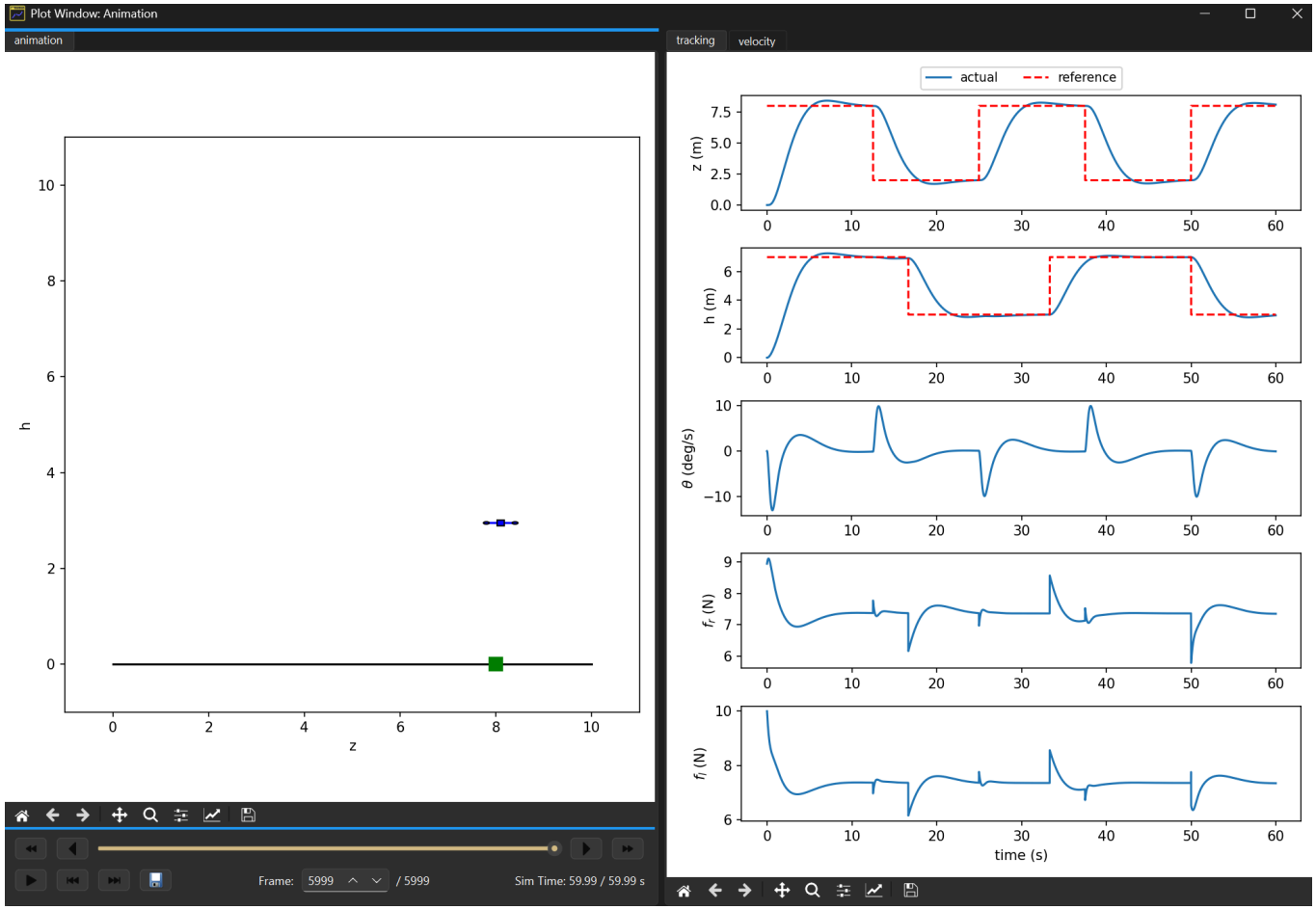

## E.11

```
des_poles: [-10.4688+10.472j  -10.4688-10.472j   -1.0469 +1.0472j  -1.0469 -1.0472j]
K: [[-25.3579 100.0362 -19.4307   8.6848]]
kr: [[-18.4979]]
```




# F.11

```
des_poles = array([-0.4487+0.4488j, -0.4487-0.4488j, -3.1406+3.1416j, -3.1406-3.1416j,
       -0.4487+0.4488j, -0.4487-0.4488j])
self.K = array([[-0.    ,  0.6041,  0.    , -0.    ,  1.346 ,  0.    ],
       [-0.0399,  0.    ,  1.2447, -0.093 , -0.    ,  0.3499]])
self.kr = array([[-0.    ,  0.6041],
       [-0.0399,  0.    ]])
```

(e) To decrease rise time we would want to move the real part of our poles more negative. We

would want to decrease the imaginary part to decrease overshoot.